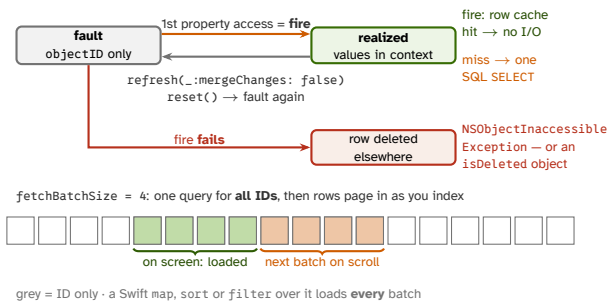


In one line: Fetched objects are **faults**, filled on first touch from the row cache or a SQL SELECT — performance = few round-trips. **Batch requests** run as SQL on the store: fast, but no context sees them until you **merge the returned IDs**. **Persistent history** logs every transaction (who, what) in the store, so each context, process and extension catches up from its own **token**.

A fault's life — and the batch-size window



Fetch knobs

Knob (default)	Effect · use
returnsObjectsAsFaults (true)	false: filled now; relationships still faults
includesPropertyValues (true)	false: IDs only, no row cache — deletes IDs first, rows in pages — long lists, FRC
fetchBatchSize (0 = all)	one query per key path, not a fault per object: kills N+1
relationshipKeyPathsForPrefetching	read-only dicts, no objects; + NSExpressionDescription = SUM/AVG in SQL
propertiesToFetch + .dictionaryResultType	SELECT COUNT · LIMIT
count(for:).fetchLimit	

A fetch **always runs SQL**; its faults then fire from the row cache it filled. Big import: save + reset() every N rows.

Batch requests — SQL on the store (SQLite only)

Request	iOS	Notes · result type for IDs
NSBatchUpdateRequest	8	propertiesToUpdate = one UPDATE .updatedObjectIDsResultType
NSBatchDeleteRequest	9	basic delete rules; Deny unsupported .resultTypeObjectIDs
NSBatchInsertRequest	13	dicts or a handler; unique constraints checked; no relationships (<i>unverified</i>) · .objectIDs

All skip the context: no validation, no willSave, no notifications; objects in memory go stale. Merge the IDs: class method mergeChanges(fromRemoteContextSave:into:) — or let history do it.

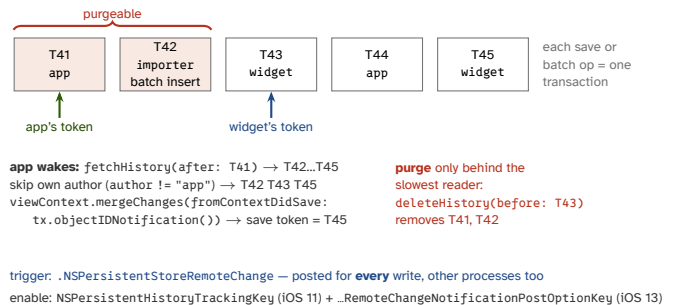
Example — FRC + diffable, and a batch delete the UI sees

```
let r = NSFetchedRequest<Item>(entityName: "Item")
r.sortDescriptors = [NSSortDescriptor(key: "date", ascending: false)]
r.fetchBatchSize = 30
frc = NSFetchedResultsController(fetchRequest: r, managedObjectContext:
    viewContext, sectionNameKeyPath: nil, cacheName: nil)
frc.delegate = self; try frc.performFetch()
func controller(_ c: NSFetchedResultsController<NSFetchRequestResult>,
    didChangeContentWith s: NSDiffableDataSourceSnapshotReference) {
    dataSource.apply(s as NSDiffableDataSourceSnapshot<String, NSManagedObjectID>)
}
// inside bg.perform: 50k rows gone in one DELETE, then tell viewContext
let del = NSBatchDeleteRequest(fetchRequest: NSFetchedRequest(entityName: "Item"))
del.resultType = .resultTypeObjectIDs
let res = try bg.execute(del) as? NSBatchDeleteResult
let ids = res?.result as? [NSManagedObjectID] ?? []
NSManagedObjectContext.mergeChanges(fromRemoteContextSave:
    [NSDeletedObjectsKey: ids], into: [viewContext]) // FRC animates it
```

NSFetchedResultsController

- Watches **one context** for one fetch request (≥ 1 sort descriptor; sectionNameKeyPath follows the first sort key). It moves only when a change lands *in that context* — a save there or a merge.
- iOS 13 controller(_:didChangeContentWith:): a snapshot of NSManagedObjectContext (sections = String);

Persistent history — the store's changelog



implement it and the per-row callbacks stop. A CollectionDifference variant exists too.

- An in-place edit keeps the same ID, so the diff sees *nothing*: reconfigureItems (iOS 15) the updated IDs, or let cells observe their object.

Persistent history

- Each NSPersistentHistoryTransaction: token, timestamp, author, contextName, changes (object ID, insert/update/delete, updatedProperties). Name writers: ctx.transactionAuthor = "importer".
- Read on a background context with NSPersistentHistoryChangeRequest.fetchHistory(after:); persist the last token (NSSecureCoding) per reader.
- Purge with deleteHistory(before:) a token, date or transaction. Once tracked, opening the store *without* the key makes it **read-only**.

App + extension on one store

Store URL in the App Group container; both targets set both keys on the store description **before** loadPersistentStores. Each process has its own coordinator, so automaticallyMergesChangesFromParent never sees the other — each reader keeps its own token in the group and merges foreign authors.

Debug — launch arguments

-com.apple.CoreData.SQLDebug 1 (up to 4): every SQL statement + time — N+1 shows as a SELECT per row.
...ConcurrencyDebug 1: crash on wrong-queue access.
...MigrationDebug 1: migration steps.

Traps

- ! FRC stale after an import: the change never reached its context (no merge, a batch request, another process).
- ! Batch delete, UI still shows the rows — nobody merged the IDs.
- ! Purging on a timer before the widget read it: it silently misses changes.
- ! Reading objectID or isFault does not fire a fault.

Remember

Faults are free until touched; batches are fast until forgotten; history is read after your token and purged behind the slowest reader.